MILITARY SPECIFICATION

PROPELLANT M9, FLAKE
(FOR USE IN 40MM CARTRIDGES)

1. SCOPE

1.1 This specification covers M9 Flake Propellant for use in 40MM cartridges (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-286	-	Propellant, Solid, Sampling, Examinations and Testing
MIL-STD-652	-	Propellant, Solid for Cannons, Requirements and Packing
MIL-STD-1168	-	Lot Numbering of Ammunition

DRAWINGS

ARMY

C-122-2-18	Head Assembly, Edgewood Arsenal
D8844609	Case, Cartridge, Loading Assembly, M118, Picatinny Arsenal
C9207988	Case, Cartridge, Loading Assembly, XM195, Picatinny Arsenal

FSC: 1376

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 49 CFR 171-179

The Interstate Commerce Commission Rules are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C., 20402. Orders for the above publication should cite "49 CFR 171-179, (latest revision)."

3. REQUIREMENTS

3.1 Materials.-The propellant shall be M9, Flake, conforming to the requirements of Standard MIL-STD-652, and Drawings (dwgs) listed in Table I, except that the graphite content shall be 0.4 percent, maximum (max.).

TABLE I

Cartridge Designation	Drawing
M118	D8844609
Head Assembly	C122-2-18
XM195	C9207988

3.2 Form.-The propellant shall be in the form of solid discs when examined as specified in 4.3.2.

3.3 Physical properties.-The physical properties shall be as specified in Table II when tested as specified in 4.3.1 and 4.3.2.

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TABLE II

Diameter, inches (advisory)	0.035
Thickness, inches (advisory)	0.00325
Bulk density lbs/cu. ft., min.	32.0

3.4 Closed Bomb, Relative Quickness and Relative Force.-The M9 propellant shall have a relative quickness and a relative force of 100 plus or minus 3 percent when tested as specified in 4.3.3.

3.5 Workmanship.-The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

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a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of batches of product inspected, (shall be made available upon request by the contracting officer).
c. Results obtained, by defect code, for all inspections performed, (shall be made available upon request by the contracting officer).
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance an all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of pounds in the lot.
g. Date submitted.
h. Eight copies of propellant description sheet AMC Form 1047 (00 Form 1204) containing the applicable data (see 6.4).

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying orgznization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous quantity of propellant from which the representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of propellant from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of one or more batches of M9 propellant which has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. Unless otherwise approved by the contracting officer, the inspection lot size of propellant deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the larges quantity contractually scheduled for delivery during any month of the contract period. The classification of defects shall be in accordance with Standard MIL-STD-105. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168.

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4.2.2 Testing

4.2.2.1 Sampling.-The sampling shall be in accordance with Standard MIL-STD-652, except that a five-pound sample shall be used for chemical and physical tests. If the sample fails to meet the requirements of paragraph 3, the lot shall be rejected.

4.3 Test methods and procedures

4.3.1 Bulk density, Major defect, Code No. 01001.-The bulk density shall be determined in accordance with Method 502.2 or 502.1 of Standard MIL-STD-286.

4.3.2 Form and dimensions, Major defect, Code No. 02001.-One hundred flakes shall be examined under a microscope for form and dimension.

4.3.3 Closed bomb test, Major defect, Code No. 03001.-The propellant shall be tested in accordance with Method 801.1 of Standard MIL-STD-286 (see 6.4 and 6.5).

4.3.4 Determination of graphite, Minor defect, Code No. 04001. The graphite content shall be determined as specified in MIL-STD-286, Method 309.1.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level C.-Propellant shall be packed in a manner that shall conform to the requirements contained in the Code of Federal Regulations, CFR 49, 171-179, for shipment of propellant, and in a manner that will insure safe delivery from manufacturer's plant to the first receiving activity.

5.2 Marking.-Marking shall be in accordance with the manufacturer's normal practice and shall conform to the Code of Federal Regulations, CFR 49, requirements for interplant shipment of propellant, and Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Process.-Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

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6.2 Inspection code numbers.-The five-digit code numbers assigned o the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Intended use.-It is intended that the item of this specification will be used in the following case cartridges: M118, XM195 and the head assembly.

6.4 Submission of AMC Form 1047.-Information contained in the propellant description sheet, AMC Form 1047, need not be resubmitted in the form of a certificate provided the provisions regarding certifying of responsible agents is adhered to (see 4.1.1). The Form 1047 shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2 and SMUPA-DE2, Dover, New Jersey, 07801.

6.5 Closed Bomb Test.-Lot No. HEP-63004 should be used as a standard in the closed bomb test.

CUSTODIAN:
ARMY-MU

PREPARING ACTIVITY:
ARMY-MU

Project Number: 1376-A025

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DEPARTMENT OF THE ARMY
PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

COMMANDING OFFICER
United States Army
Picatinny Arsenal
ATTN: SMUPA-ND
Dover, New Jersey 07801

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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